L26 28/10/2020

Let $\gamma > 0$ and

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RL}_\infty \quad \text{and} \quad \sigma_{max}(D) < \gamma$$

Then $\|G\|_\infty < \gamma$ iff H has no imaginary axis eigenvalues, where

$$H := \begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix} \quad \text{and} \quad R := \gamma^2 I - D^*D$$

$A \to \{\sigma_1, \sigma_2 \dots \sigma_r\}$

$\boxed{\sigma_i^2} = \underline{\lambda_i(A^TA)}$

Proof: $\Phi(s) := \gamma^2 I - G^T(-s)\,G(s)$

$$\|G\|_\infty^2 < \gamma^2 \iff \sigma_i^2(G(j\omega)) < \gamma^2 \quad \forall\, \omega \in \mathbb{R}$$

$\lambda \in \text{spec}(P)$

$\gamma - \lambda \in \text{spec}(\gamma I - P)$

$$\iff \lambda_i\left(G^T(-j\omega)G(j\omega)\right) < \gamma^2 \quad \forall\, \omega \in \mathbb{R}$$

$$\iff \lambda_i\left(\gamma^2 I - G^T(-j\omega)G(j\omega)\right) > 0 \quad \forall\, \omega \in \mathbb{R}$$

$$\iff \Phi(j\omega) > 0 \quad \forall\, \omega \in \mathbb{R}$$

$$\Updownarrow$$

$\Phi(j\omega)$ is non-singular for all $\omega \in \mathbb{R} \cup \{\infty\}$

$\boxed{M(s)}$

$$\Updownarrow$$

$\Phi(s)$ has no imaginary axis zeros. $\iff$ $\Phi^{-1}(s)$ has no $j\mathbb{R}$ poles

$\|G\|_\infty < \gamma \iff \Phi^{-1}(s)$ has no imaginary axis poles $\iff \text{spec}(H) \cap j\mathbb{R} = \phi$

H (Hamiltonian matrix)

$$\Phi^{-1}(s) = \left[\begin{array}{cc|c} A + BR^{-1}D^*C & BR^{-}B^* & BR^{-1} \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* & -C^*DR^{-1} \\ \hline R^{-1}D^*C & R^{-1}B^* & R^{-1} \end{array}\right]$$

Let $j\omega_0 \in \text{spec}(H)$ but $j\omega_0 \notin$ poles of $\Phi^{-1}(s)$

Case I: $j\omega_0$ is an unobservable mode $\left(\begin{bmatrix} R^{-1}D^*C & R^{-1}B^* \end{bmatrix}, H\right)$

Case II: $j\omega_0$ is uncontrollable

$\exists x$ of suitable dimension such that

λ (C, A) $\exists v$ s.t. $\begin{bmatrix} A-\lambda I \\ C \end{bmatrix} v = 0$

$$\begin{bmatrix} H - j\omega_0 I \\ R^{-1}D^*C \quad R^{-1}B^* \end{bmatrix} x = 0 \Rightarrow (H - j\omega_0 I)x = 0$$

$$\begin{bmatrix} R^{-1}D^*C & R^{-1}B^* \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0, \text{ where } x := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow R^{-1}D^*C x_1 + R^{-1}B^* x_2 = 0 \Rightarrow D^*Cx_1 + B^* x_2 = 0$$

$$\begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = j\omega_0 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(A + BR^{-1}D^*C)x_1 + BR^{-1}B^* x_2 = j\omega_0 x_1 \Rightarrow (A - j\omega_0 I)x_1 + BR^{-1}(\cancelto{0}{D^*Cx_1} + B^* x_2) = 0$$

$$\therefore (A - j\omega_0 I)x_1 = 0 \Rightarrow j\omega_0 \in \text{spec}(A)$$

A has an eigenvalue on $j\mathbb{R}$ $\boxed{\text{not possible}}$

$$\therefore x_1 = 0$$

$$-[C^*(I + DR^{-1}D^*)C]x_1 - [(A + BR^{-1}D^*C)^*]x_2 = j\omega_0 x_2$$

$$\Rightarrow -\cancel{C^*Cx_1} - \cancelto{0}{C^*DR^{-1}D^*Cx_1} - A^*x_2 - \cancelto{0}{C^*DR^{-1}B^*x_2} = j\omega_0 x_2 \Rightarrow (j\omega_0 I + A^*)x_2 = 0$$

$j\omega_0 \in \text{spec}(A^*)$

$\Rightarrow -j\omega_0 \in \text{spec}(A)$ $\boxed{\text{not possible}}$

$$\therefore x_2 = 0$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$\therefore$ $j\omega_0 \in \text{spec}(H)$ then it is contr & obs.

$\|G\|_\infty < \gamma \iff \Phi^{-1}(s)$ has no imaginar axis poles $\iff \text{spec}(H) \cap j\mathbb{R} = \phi$

Algorithm to compute $\mathcal{L}_\infty$ norms:

Let γ_{Lb} and γ_{ub} be some lower & upper bounds on $\|G\|_\infty$.

$$\underline{\underline{\gamma_{Lb}}} < \|G\|_\infty < \underline{\underline{\gamma_{ub}}}$$

$$\gamma_{Lb} = \max\left\{\sigma_{max}(D),\ \sqrt{\mathrm{Trace}(W_c W_o)/n}\right\}$$

$$\gamma_{ub} = \sigma_{max}(D) + 2\sqrt{n\,\mathrm{Trace}(W_c W_o)}$$

Steps:

$\gamma_L := \gamma_{Lb}$

$\gamma_u := \gamma_{ub}$

repeat {

$$\gamma := \frac{\gamma_L + \gamma_u}{2}$$

Form H

if $\sigma(H) \cap j\mathbb{R} = \phi$, $\gamma_H := \gamma$

→ else $\gamma_L := \gamma$

} until $\{\gamma_H - \gamma_L \leq \underline{\underline{2\varepsilon\gamma_L}}\}$ ← ε tolerance

$$\|G\|_\infty \approx \left(\frac{\gamma_L + \gamma_H}{2}\right)$$

Error in $\mathcal{L}_\infty$-norm comp:

$$\left(\frac{\gamma_L + \gamma_H}{2}\right) - \|H\|_\infty \leq \varepsilon \|H\|_\infty$$